Homework 6

《生物医学图像处理》

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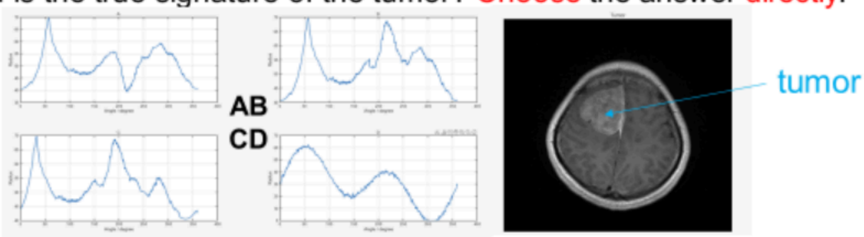
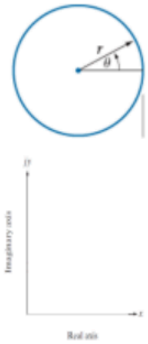
HW1- Signature and Fourier Descriptors

1.Which figure in is the true signature of the tumor? Choose the answer directly.





2.Plot the rectangular coordinate $f(x,y)$of the $r(\theta)$ (no need to derive $f(x,y)$)

$$r(\theta) = 2 - 2sin\theta + \frac{sin\theta\sqrt{|cos\theta|}}{sin\theta + 1.5}, \theta = \frac{2\pi n}{1280}, n = 0,1,2 \ldots 1279$$

3.Along the $f(x,y)$ boundary, get Fourier Descriptors. Set all $a(u) = 0$ except $a(0), a(1)$. Plot the restored boundary and derive its equation. You can use matlab or python to help you calculate its coefficient. Keep two decimal places.

4.Replace $r(\theta)$ with $\rho(\theta) = \frac{1}{|\cos(\theta)|+|\sin(\theta)|}, \theta = \frac{2\pi n}{1280}, n = 0,1,2 \ldots 1279$, redo 2 and 3.

(***In your figures, student number as the watermark***).

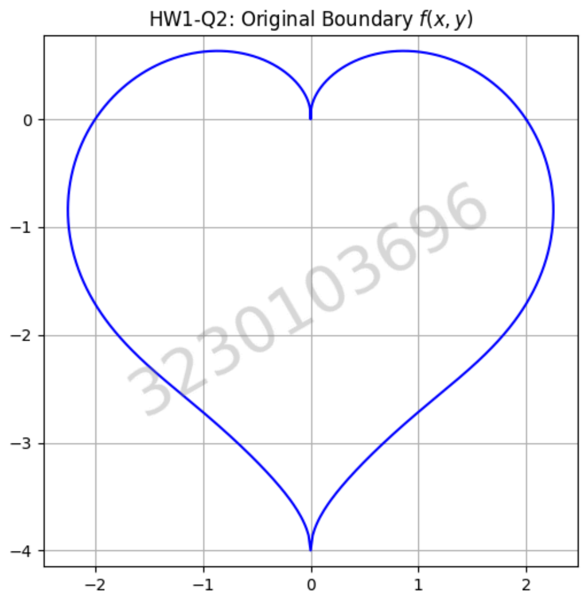
SOLUTION:

(1) **A**.

(2) python 代码如下：

```python
N = 1280
n = np.arange(N)
theta = 2 * np.pi * n / N
student_id = "3230103696"

def add_watermark():
    """在图像中心添加斜向半透明的水印"""
    plt.text(0.5, 0.5, student_id, transform=plt.gca().transAxes,
             fontsize=40, color='gray', alpha=0.3,
             ha='center', va='center', rotation=30)

# ================= Q2 & Q3 =================
# 定义原始的 r(theta)
r_theta = 2 - 2 * np.sin(theta) + (np.sin(theta) * np.sqrt(np.abs(np.cos(theta)))) / (np.sin(theta) + 1.5)
x = r_theta * np.cos(theta)
y = r_theta * np.sin(theta)

# 绘制原边界 (Q2)
plt.figure(figsize=(6, 6))
plt.plot(x, y, 'b')
plt.title("HW1-Q2: Original Boundary $f(x,y)$")
plt.axis('equal')
plt.grid(True)
add_watermark()
```

```python
plt.show()

# 计算傅里叶描述子
s_n = x + 1j * y
A_u = np.fft.fft(s_n) / N  # 除以 N 才是标准数学定义下的系数值

# 提取并打印系数，保留两位小数
a0 = np.round(A_u[0], 2)
a1 = np.round(A_u[1], 2)
print(f"--> 第一部分方程的系数: a(0) = {a0}, a(1) = {a1}")

# 滤波: 只保留 a(0) 和 a(1)
A_u_filtered = np.zeros_like(A_u)
A_u_filtered[0] = A_u[0]
A_u_filtered[1] = A_u[1]

# 逆变换还原边界（乘以 N 恢复幅度）
s_restored = np.fft.ifft(A_u_filtered * N)

# 绘制恢复后的边界（Q3）
plt.figure(figsize=(6, 6))
plt.plot(np.real(s_restored), np.imag(s_restored), 'r')
plt.title("HW1-Q3: Restored Boundary (only $a(0), a(1)$)")
plt.axis('equal')
plt.grid(True)
add_watermark()
plt.show()
```



(3) 首先，将离散的边界坐标表示为复平面上的离散序列：

$$s(n) = x(n) + jy(n) = r(\theta_n)e^{j\theta_n}$$

对其进行离散傅里叶变换（获取傅里叶描述子）：

$$a(u) = \frac{1}{N}\sum_{n=0}^{N-1} s(n)e^{\frac{-j2\pi un}{N}}, \quad u = 0, 1, ..., N-1$$

根据题目要求，将除 $a(0)$ 和 $a(1)$ 之外的所有高频系数 $a(u)$ 设为 0。随后通过逆离散傅里叶变换重构平滑后的边界：

$$s'(n) = a(0) + a(1)e^{\frac{j2\pi n}{N}} = a(0) + a(1)e^{j\theta_n}, \theta_n = \frac{2\pi n}{1280}$$

令 $a(0) = A_0 + jB_0$，$a(1) = A_1 + jB_1$。利用欧拉公式 $e^{j\theta_n} = \cos(\theta_n) + j\sin(\theta_n)$，将复数方程拆解为直角坐标系下的实部和虚部：

$$\begin{aligned} x'(n) + jy'(n) &= (A_0 + jB_0) + (A_1 + jB_1)(\cos\theta_n + j\sin\theta_n) \\ &= (A_0 + A_1\cos\theta_n - B_1\sin\theta_n) + j(B_0 + A_1\sin\theta_n + B_1\cos\theta_n) \end{aligned}$$

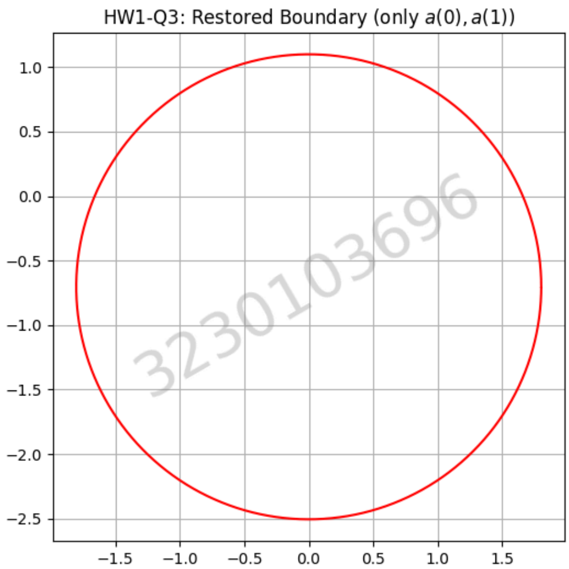
通过 python 代码计算得$a(0) = -0.7j, a(1) = 1.8$

$$x'(n) = 1.8\cos\theta_n$$

$$y'(n) = -0.7 + 1.8\sin\theta_n$$

因此方程为

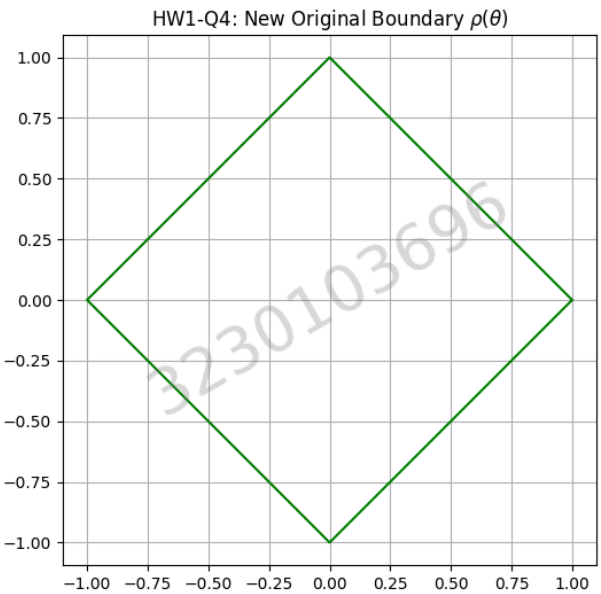
$$x^2 + (y + 0.7)^2 = 1.8^2$$



(4) python 代码如下：

```
# ================= Q4 =================
# 定义新的 rho(theta)
rho_theta = 1 / (np.abs(np.cos(theta)) + np.abs(np.sin(theta)))
x_rho = rho_theta * np.cos(theta)
y_rho = rho_theta * np.sin(theta)

# 绘制新原边界 (Q4 - Redo 2)
plt.figure(figsize=(6, 6))
plt.plot(x_rho, y_rho, 'g')
plt.title("HW1-Q4: New Original Boundary $\\rho(\\theta)$")
plt.axis('equal')
plt.grid(True)
add_watermark()
```

```python
plt.show()

# 计算新的傅里叶描述子
s_n_rho = x_rho + 1j * y_rho
A_u_rho = np.fft.fft(s_n_rho) / N

a0_rho = np.round(A_u_rho[0], 2)
a1_rho = np.round(A_u_rho[1], 2)
print(f"--> 第二部分方程的系数: a(0) = {a0_rho}, a(1) = {a1_rho}")

# 滤波与逆变换
A_u_rho_filtered = np.zeros_like(A_u_rho)
A_u_rho_filtered[0] = A_u_rho[0]
A_u_rho_filtered[1] = A_u_rho[1]
s_restored_rho = np.fft.ifft(A_u_rho_filtered * N)

# 绘制新恢复边界 (Q4 - Redo 3)
plt.figure(figsize=(6, 6))
plt.plot(np.real(s_restored_rho), np.imag(s_restored_rho), 'm')
plt.title("HW1-Q4: Restored New Boundary")
plt.axis('equal')
plt.grid(True)
add_watermark()
plt.show()
```



同理，计算得：

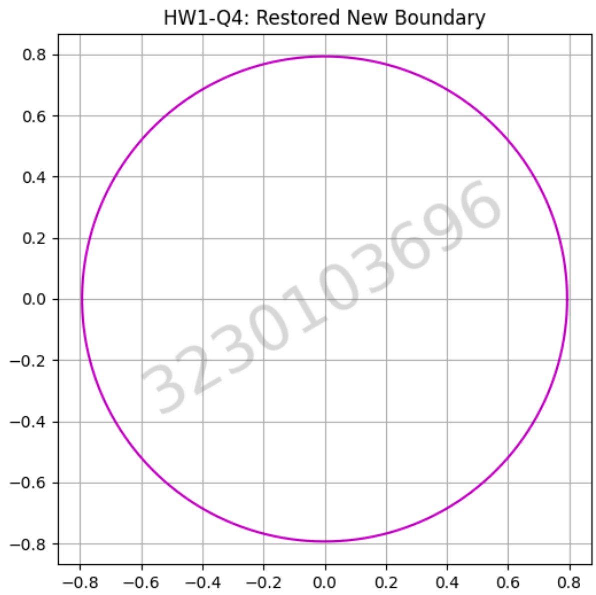
$$a(0) = 0, a(1) = 0.79$$

$$x'(n) = 0.79 \cos \theta_n$$

$$y'(n) = 0.79 \sin \theta_n$$

因此方程为

$$x^2 + y^2 = 0.79^2$$


HW1-Q4: Restored New Boundary
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8
3230103696

2 HW2 - First Order Features

(1) Given the image I_0, carefully calculate its energy, range, μ_1, μ_2, μ_3, uniformity and entropy.

$$I_0 = \begin{bmatrix} 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 3 & 1 \\ 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & 1 \end{bmatrix}$$

(2) Assume that the image is a tiny temperature map on a world whose boundaries wrap around. The process of heat diffusion is analogous to applying circular convolution to an image. Using circular padding:

$$I_{\{t+1\}} = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} *_{\text{circular}} I_t$$

Which statement best describes this diffusion process from the perspective of images in practice?

- A. It gently softens into the twilight, gradually losing its forms as its total brightness dissipates into the surrounding void.
- B. As pixels endlessly intertwine in the cycle, the image awakens, stretching its dynamic range to forge sharper and deeper boundaries.
- C. It does not perish in an instant; it slowly forgets its local forms, preserving its total brightness as a faint memory of the past, until all pixels drift toward the same quiet average — and the final whispers vanish into rounding and noise.
- D. Like a drop of ink blooming in a calm pond, the image embraces absolute smoothness by continuously breathing new information into its canvas.

(3) Fill in the table for I_∞ directly.

	energy	**range**	μ_1	μ_2	μ_3	**uniformity**	**entropy**
I_∞							

SOLUTION:

(1)

首先统计 I_0 中各灰度值的出现频次与概率 $p(z)$：

- 灰度 0：出现 10 次，$p(0) = \frac{10}{25} = 0.4$
- 灰度 1：出现 8 次，$p(1) = \frac{8}{25} = 0.32$
- 灰度 2：出现 4 次，$p(2) = \frac{4}{25} = 0.16$
- 灰度 3：出现 3 次，$p(3) = \frac{3}{25} = 0.12$

根据定义公式进行计算：

- **能量 (Energy)**:
  $E = \sum I_0^2 = 51$
- **极差 (Range)**:
  $\max(z) - \min(z) = 3 - 0 = 3$
- **均值 (μ_1)**:
  $\mu_1 = \sum z_i p(z_i) = 1.0$
- **方差 (μ_2)**:
  $\mu_2 = \sum (z_i - \mu_1)^2 p(z_i) = 1.04$
- **偏度 (μ_3)**:
  $\mu_3 = \sum (z_i - \mu_1)^3 p(z_i) = 0.72$
- **均匀度 (Uniformity)**:
  $U = \sum (p(z_i))^2 = 0.4^2 + 0.32^2 + 0.16^2 + 0.12^2 = 0.3024$
- **熵 (Entropy)**:
  $e = -\sum p(z_i) \log_2 p(z_i) = -(0.4 \log_2 0.4 + 0.32 \log_2 0.32 + 0.16 \log_2 0.16 + 0.12 \log_2 0.12) \approx 1.8449$

(2) **C**.

相当于每一步都把一个像素变成周围 3×3 邻域的平均值。由于使用 circular padding，边界也是首尾相接的，所以整个图像的总亮度不会改变。

不断扩散之后，局部差异会被平均掉，最后所有像素都会趋向于同一个值，也就是 原图像的全局平均值。

(3)

Image	energy	range	μ_1	μ_2	μ_3	uniformity	entropy
I_∞	25	0	1.0	0	0	1	0

3 HW3 - GLCM Find the co-occurrence matrix of a matrix pattern in the following cases.

(1) The position operator Q is defined as "one pixel to the right"

(2) The position operator Q is defined as "two pixels to the right"

(3) For 1. and 2.'s GLCM, calculate contrast and homogeneity.

$$M = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 \end{bmatrix}$$

SOLUTION:

(1)

逐行统计相邻像素对 (i, j)，5 行共 20 对。计数矩阵 M_1 为：

$$M_1 = \begin{bmatrix} 0 & 3 & 0 \\ 3 & 0 & 7 \\ 0 & 7 & 0 \end{bmatrix}$$

归一化后得到共生矩阵 $P_1 = \frac{M_1}{20}$：

$$P_1 = \begin{bmatrix} 0 & 0.15 & 0 \\ 0.15 & 0 & 0.35 \\ 0 & 0.35 & 0 \end{bmatrix}$$

(2)

逐行统计间隔一个像素的组合对 (i, j)，5 行共 15 对。计数矩阵 M_2 为：

$$M_2 = \begin{bmatrix} 0 & 0 & 3 \\ 0 & 7 & 0 \\ 3 & 0 & 2 \end{bmatrix}$$

归一化后得到共生矩阵 $P_2 = \frac{M_2}{15}$：

$$P_2 = \begin{bmatrix} 0 & 0 & \frac{1}{5} \\ 0 & \frac{7}{15} & 0 \\ \frac{1}{5} & 0 & \frac{2}{15} \end{bmatrix}$$

(3)

$$\text{对比度}: \sum_{i,j} (i-j)^2 P(i,j)$$

$$\text{同质性}: \sum_{i,j} \frac{P(i,j)}{1+|i-j|}$$

GLCM 1 (P_1):

所有非零项 (i,j) 均满足 $|i-j|=1$：

- 对比度 ： $1^2 \times (0.15+0.15+0.35+0.35) = 1.0$
- 同质性 ： $\frac{0.15}{1+1}+\frac{0.15}{1+1}+\frac{0.35}{1+1}+\frac{0.35}{1+1}=\frac{1.0}{2}=0.5$

GLCM 2 (P_2):

非零项包括 $(0,2),(2,0)$和 $(1,1),(2,2)$：

- 对比度 ： $=4\times\left(\frac{1}{5}+\frac{1}{5}\right)+0\times\left(\frac{7}{15}+\frac{2}{15}\right)=1.6$
- 同质性 ： $=\frac{\frac{1}{5}}{1+2}+\frac{\frac{1}{5}}{1+2}+\frac{\frac{7}{15}}{1+0}+\frac{\frac{2}{15}}{1+0}=\frac{0.4}{5}+\frac{9}{15}\approx 0.73$